matlab.mathworks.com

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KARTHIK

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FILE

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LABELS AND ANNOTATIONS

/ > MATLAB Drive >

Files

Workspace

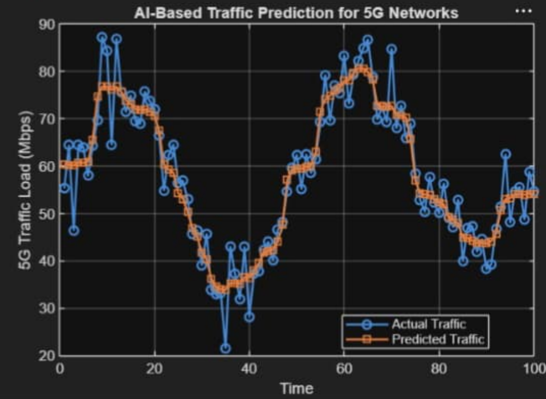
Name	Value	Size	Class
model	1×1 Regressi...	1×1	classreg.learn...
RMSE	4.4915	1×1	double
t	100×1 double	100×1	double
traffic	100×1 double	100×1	double
X	100×1 double	100×1	double
Y	100×1 double	100×1	double
Y_pred	100×1 double	100×1	double

assesment.m | ex11.m | ex12.m | ex13.m | ex14.m | ex15.m

/MATLAB Drive/ex15.m

```
model = fitrensemble(X,Y,'Method','Bag');

% Predict traffic
Y_pred = predict(model,X);

% Calculate RMSE
RMSE = sqrt(mean((Y-Y_pred).^2));

% Plot actual and predicted traffic
figure;
plot(t,Y,'o-','LineWidth',1.5);
hold on;
plot(t,Y_pred,'s-','LineWidth',1.5);

xlabel('Time');
ylabel('5G Traffic Load (Mbps)');
title('AI-Based Traffic Prediction for 5G Networks');
legend('Actual Traffic','Predicted Traffic','Location','best'
grid on;
```

Figure 1



Command Window

```
Prediction RMSE = 4.49 Mbps
>>
```

36°C
Partly sunny

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ENG
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14:40
17-08-2026